

Lab 1: Inspecting Table Size

The screenshot shows a PostgreSQL client interface with a query editor and a results pane. The query editor contains the following SQL code:

```
-- 1. Check table vs index size
SELECT
  relname AS object_name,
  pg_size_pretty(pg_table_size(fed.oid)) AS data_size,
  pg_size_pretty(pg_indexes_size(fed.oid)) AS index_size,
  pg_size_pretty(pg_total_relation_size(fed.oid)) AS total_size
FROM pg_class fed
JOIN pg_namespace n ON n.oid = fed.relnamespace
WHERE fed.relkind = 'r' AND n.nspname = 'public'
ORDER BY pg_total_relation_size(fed.oid) DESC;
```

The results pane displays a table with the following data:

object_name	data_size	index_size	total_size
rental	1232 kB	1120 kB	2352 kB
payment	896 kB	920 kB	1816 kB
film	736 kB	200 kB	936 kB
film_actor	272 kB	216 kB	488 kB
inventory	232 kB	208 kB	440 kB
customer	96 kB	112 kB	208 kB
address	88 kB	64 kB	152 kB
film_category	72 kB	40 kB	112 kB
city	64 kB	48 kB	112 kB

Total rows: 17 | Query complete 00:00:01.121 | CRLF | Ln 10, Col 47

Lab 2: The Impact of a New Index

1. Step 1:

The screenshot shows a PostgreSQL client interface with a query editor and a results pane. The query editor contains the following SQL code:

```
-- 2. The impact of a New Index
-- Step 1: Check current size of 'payment' indexes
SELECT pg_size_pretty(pg_indexes_size('payment'));
```

The results pane displays a table with the following data:

pg_size_pretty
920 kB

Showing rows: 1 to 1 | Page No: 1 of 1

2. Step 2:

Query

Query History

1

-- 2. The impact of a New Index

2

-- Step 1: Check current size of 'payment' indexes

3

SELECT pg_size_pretty(pg_indexes_size('payment'));

4

5

-- Step 2: Create a heavy index on payment_date

6

CREATE INDEX idx_payment_date_long ON payment (payment_date, amount, customer_id);

Data Output

Messages

Notifications

CREATE INDEX

Query returned successfully in 133 msec.

3. Step 3:

Query

Query History

1

-- 2. The impact of a New Index

2

-- Step 1: Check current size of 'payment' indexes

3

SELECT pg_size_pretty(pg_indexes_size('payment'));

4

5

-- Step 2: Create a heavy index on payment_date

6

CREATE INDEX idx_payment_date_long ON payment (payment_date, amount, customer_id);

7

8

-- Step 3: Check size again

9

SELECT pg_size_pretty(pg_indexes_size('payment'));

Data Output

Messages

Notifications

pg_size_pretty

text

1

1488 kB

Showing rows: 1 to 1 Page No: 1 of 1

4. Discussion: How many Pages (8KB) did this new index consume?

>> $1488 : 8 = 186$ Pages

Lab 3: Partial Index Practice

1. Create index

Query Query History

```
1 -- 3. Partial Index Prectice
2 -- Create Partial Index
3 CREATE INDEX idx_expensive_films
4 ON film (replacement_cost)
5 WHERE replacement_cost > 25.00;
```

Data Output Messages Notifications

CREATE INDEX

Query returned successfully in 107 msec.

2. Find the size of this specific index

Query Query History

```
1 -- 3. Partial Index Prectice
2 -- Create Partial Index
3 CREATE INDEX idx_expensive_films
4 ON film (replacement_cost)
5 WHERE replacement_cost > 25.00;
6
7 -- Find the size of this specific index
8 SELECT pg_size_pretty(pg_relation_size('idx_expensive_films')) AS partial_index_size;
```

Data Output Messages Notifications

Showing rows: 1 to 1 Page No: 1

	partial_index_size
1	16 kB

3. Create a full index

Query

Query History

```

1  -- 3. Partial Index Prectice
2  -- Create Partial Index
3  CREATE INDEX idx_expensive_films
4  ON film (replacement_cost)
5  WHERE replacement_cost > 25.00;
6
7  -- Find the size of this specific index
8  SELECT pg_size_pretty(pg_relation_size('idx_expensive_films')) AS partial_index_size;
9
10 -- Create a full index
11 CREATE INDEX idx_full_replacement_cost
12 ON film (replacement_cost);

```

Data Output

Messages

Notifications

CREATE INDEX

Query returned successfully in 88 msec.

4. Đo size của full index

Query

Query History

```

1  -- 3. Partial Index Prectice
2  -- Create Partial Index
3  CREATE INDEX idx_expensive_films
4  ON film (replacement_cost)
5  WHERE replacement_cost > 25.00;
6
7  -- Find the size of this specific index
8  SELECT pg_size_pretty(pg_relation_size('idx_expensive_films')) AS partial_index_size;
9
10 -- Create a full index
11 CREATE INDEX idx_full_replacement_cost
12 ON film (replacement_cost);
13
14 -- Đo size của full index
15 SELECT pg_size_pretty(pg_relation_size('idx_full_replacement_cost')) AS full_index_size;

```

Data Output

Messages

Notifications

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SQL

Showing rows: 1 to 1 Page No: 1 of 1

	full_index_size	text
1	40 kB	